Homework 2. Due by Tuesday February 10 at 11:59 PM.

Please copy all your queries (sql code) from MySQL Workbench and paste them into this Word document for each question. Upload this Word file with your answers to the HW2 assignment folder on Blackboard. Check the file after uploading it.

Please create a screenshot of each query result and paste after each query.

Video 3 describes the database create_safety.

The create_safety.sql script creates a database which contains the 6 tables described below. The data contains safety-related information for 179 company branch locations. To test your queries, create those tables using the create_safety.sql script.

Locations	**Field Description**
Location_ID (pk)	4 digit location id number
Headcount	Number of employees at locations
Division	2 character division code
State	Location state
City	Location city
Employee_Safety_Committee	Yes/No indicator of safety committee at location

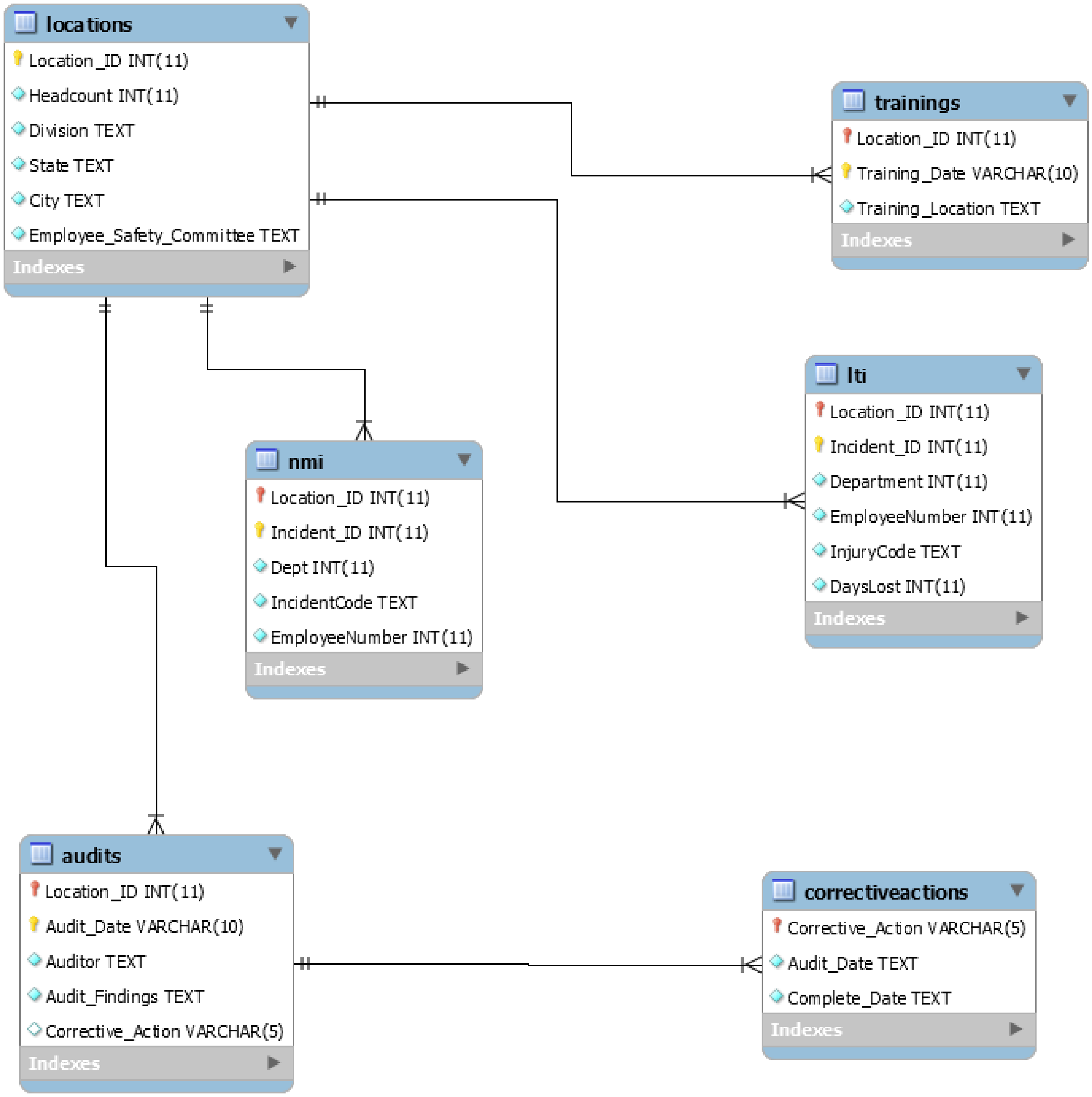
Audits	**Field Description**
Location_ID (pk and fk)	4 digit location id number
Audit_Date (pk)	Date of audit
Auditor	Name of head auditor
Audit_Findings	Pass/Fail indicator of audit results
Corrective_Action (Unique)	5 digit CA number, if any (default = NULL)

Correctiveactions	**Field Description**
Corrective_Action (pk and fk)	5 digit CA number, if any (default = NULL)
Audit_Date	Date of audit
Complete_Date	Date CA was completed

Trainings	**Field Description**
Location_ID (pk and fk)	4 digit location id number
Training_Date (pk)	Date of training
Training_Location	Location of training (Onsite/Offsite)

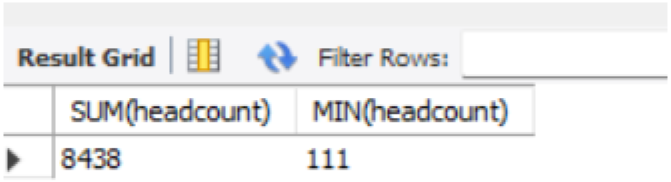
lti (Lost Time Incidents)	**Field Description**
Location_ID (pk and fk)	4 digit location id number
Incident_ID (pk)	4 digit incident id number
Department	2 digit department number
EmployeeNumber	5 digit employee number
InjuryCode	Description of injury type
DaysLost	# of days lost due to injury

nmi (Near Miss Incidents)	**Field Description**
Location_ID (pk and fk)	4 digit location id number
Incident_ID (pk)	4 digit incident id number
Dept	2 digits department number
IncidentCode	Description of incident type (root cause, etc.)
EmployeeNumber	5 digit employee number



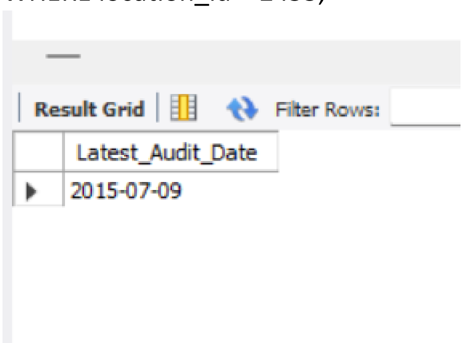
1. **(10 points total)**. What is the total headcount for division "PK" (across all locations) and minimum headcount (i.e., headcount for one location with the smallest number of people) for division "PK"? (hint: use locations table). The query needs to produce the total headcount and minimum value.

```
SELECT
   SUM(headcount),
   MIN(headcount)
FROM locations
WHERE division = 'PK';
```



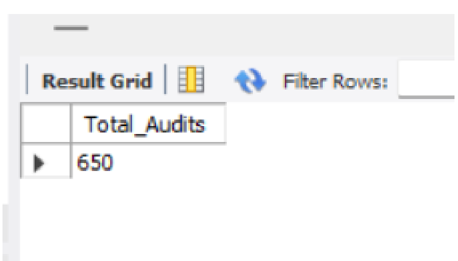
2. **(10 points total).** Show the latest audit date for location 2453. USE STR_TO_DATE function (as I show in Panopto video 3) to work with dates

```
SELECT
   MAX(STR_TO_DATE(Audit_Date, '%m/%d/%Y')) AS Latest_Audit_Date
FROM audits
WHERE location_id = 2453;
```



3. **(10 points total).** How many audits did the auditor make whose name does not include “jai” at the beginning of the name? The query needs to produce the total count of audits

```
SELECT
COUNT(*) AS Total_Audits
FROM audits
WHERE auditor NOT LIKE 'jai%';
```



4. **(10 points total)**. What is the total number of audits for each auditor whose name ends with “s”?

```
SELECT auditor, COUNT(*) AS Total_Audits
FROM audits
WHERE auditor LIKE '%s'
GROUP BY auditor;
```

Result Grid | Filter Rows:

auditor	Total_Audits
Spryos	63
DeSantis	66

5. **(10 points total)**. How many audits DID NOT HAVE corrective actions?

```
SELECT
  COUNT(*) AS Audits_Without_Corrective_Action
FROM audits
WHERE Corrective_Action IS NULL;
```

Result Grid | Filter Rows:

Audits_Without_Corrective_Action
513

6. **(10 points total)**. Use only NMI table. Count NMI incidents by location id and show only results for locations where the number of incidents is more than or equal to 21.

```
SELECT  location_id,  COUNT(*) AS Total_NMI_Incidents
FROM nmi
GROUP BY location_id
HAVING COUNT(*) >= 21;
```

Result Grid | Filter Rows:

location_id	Total_NMI_Incidents
2408	81
2415	80
2417	76
2440	79
2453	99
2464	81
2468	105
2493	110
2505	99
2528	63
2534	99

7. **(10 points total)**. Use only LTI table. Count LTI incidents by location id and show only results for locations where location_id ends with the number 8 and has the number 4 anywhere. You may use WHERE or HAVING for that.

```
SELECT
   location_id,
   COUNT(*) AS Total_LTI_Incidents
FROM lti
WHERE location_id LIKE '%4%'
AND location_id LIKE '%8'
GROUP BY location_id;
```

Result Grid | Filter Rows:

location_id	Total_LTI_Incidents
2408	37
2468	43
4118	31
4218	45
4278	32
4298	44
4368	45
4448	54
4588	43
4598	40

8. **(10 points total)**. USE Subquery. Use only 'locations' table. Show headcount by location id and show only results for locations where headcount is more than or equal to the average headcount for all locations. You may use WHERE or HAVING to show that report. Think about whether you need to use GROUP BY or not. Sort results by headcount from largest to smallest.

```
SELECT location_id, headcount
FROM locations
WHERE headcount >= (
   SELECT AVG(headcount)
   FROM locations
)
ORDER BY headcount DESC;
```

Result Grid | Filter Rows:

location_id	headcount
2759	606
4083	604
3137	604
4516	596
3776	595
3818	595
4193	590
4448	586
4636	584
2612	580
2751	574
4130	573
3734	568
2922	567

9. **(10 points total)**. Create a report (i.e., show a query) showing the number of LTI (Lost Time Incidents) by location as well as LTI Percent (LTI / Location Headcount). Only show results where LTI Percent is more 25% (i.e., more than 0.25). Also, sort the results from smallest to largest LTI Percent and show next top 5 results only after skipping top 3 (skip first 3). To be able to confirm the answer by looking at the query results, you need to group by Location_ID, Division, Headcount and add the same attributes in SELECT part.

```
SELECT  l.location_id, l.division,  l.headcount,
   COUNT(t.incident_id) AS Total_LTI,
   (COUNT(t.incident_id) / l.headcount) AS LTI_Percent
FROM locations l
JOIN lti t
   ON l.location_id = t.location_id
GROUP BY  l.location_id, l.division,  l.headcount
HAVING (COUNT(t.incident_id) / l.headcount) > 0.25
ORDER BY LTI_Percent ASC
LIMIT 5 OFFSET 3;
```

Result Grid | Filter Rows: | Export: | Wrap

location_id	division	headcount	Total_LTI	LTI_Percent
2689	AE	148	39	0.2635
2505	FD	159	42	0.2642
4309	AE	112	30	0.2679
4283	CF	192	53	0.2760
3271	AE	120	36	0.3000

10. **(10 points total)**. Create a report (i.e., show a query) showing the number of NMI (Near Miss Incidents) by location as well as NMI Percent (NMI / Location Headcount). Only show results where NMI Percent is more 25% (i.e., more than 0.25). Also, sort the results from largest to smallest NMI Percent and show next top 6 results only after skipping top 4 (skip first 4). To be able to confirm the answer by looking at the query results, you need to group by Location_ID, Division, Headcount and add the same attributes in SELECT part. Only show results for the division "PK" (you may use WHERE or HAVING to filter the results).

```
SELECT l.location_id, l.division, l.headcount, COUNT(n.incident_id) AS Total_NMI,
   (COUNT(n.incident_id) * 1.0 / l.headcount) AS NMI_Percent
FROM locations l
JOIN nmi n ON l.location_id = n.location_id
WHERE l.division = 'PK'
GROUP BY l.location_id, l.division, l.headcount
HAVING (COUNT(n.incident_id) * 1.0 / l.headcount) > 0.25
ORDER BY NMI_Percent DESC
LIMIT 6 OFFSET 4;
```

Result Grid | Filter Rows: | Export: | Wr

location_id	division	headcount	Total_NMI	NMI_Percent
3095	PK	272	106	0.38971
3534	PK	229	85	0.37118
3202	PK	191	65	0.34031
4800	PK	384	121	0.31510
2468	PK	365	105	0.28767
3973	PK	343	95	0.27697